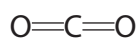


3 The following equation summarises photosynthesis.

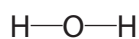


The diagrams show the bonds in carbon dioxide, water and oxygen.

carbon dioxide



water



oxygen



Energy is needed to break chemical bonds, and to form new chemical bonds. This is called the bond energy.

The table shows some bond energies for the bonds in carbon dioxide, water, glucose and oxygen.

Type of bond	Bond energy / kJ per bond
C=O	785
O—H	462
O=O	487

(a) (i) In photolysis, one of the bonds in each water molecule is broken.

Using the equation for photosynthesis, calculate how much energy is released by photolysis in order for one molecule of glucose to be made.

(1)

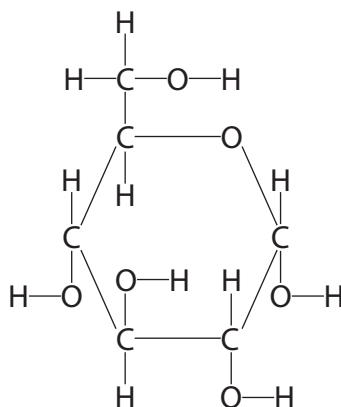
Answer kJ

(ii) Explain how light energy is converted into chemical energy in the form of ATP.

(4)

[illegible]

(b) The diagram shows the structure of glucose.



The energy needed to form the bonds in one molecule of glucose is 9164 kJ.

- (i) State what other information is needed in the table of bond energies for this value to be calculated.

(1)

- (ii) Where in the chloroplasts are these bonds formed?

(1)

- ☐ A cytoplasm
- ☐ B matrix
- ☐ C stroma
- ☐ D thylakoid membrane

(c) Glucose is used by plants in the production of amino acids.

(i) Which row of the table describes how two amino acids join together?

(1)

	Bond formed between	Type of reaction
☐ A	carbon and nitrogen	condensation
☐ B	carbon and nitrogen	hydrolysis
☐ C	oxygen and nitrogen	condensation
☐ D	oxygen and nitrogen	hydrolysis

(ii) Explain why amino acids cannot be produced from glucose alone.

(2)

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(Total for Question 3 = 10 marks)